

Software Requirement Specification (SRS)

for Student Clubs Event Management Platform

Problem Background

The primary purpose of this platform is to enable student club coordinators to post requests for scheduling events in an academic institute, which would be subject to approval by an events coordinator. Post this approval, the event would be added to the platform for registration by students. The platform would also have dedicated pages for the clubs, which the respective club's coordinator would manage.

Stakeholders / Users

The primary users of the platform are (i) students of the university, (ii) the coordinators of various clubs, (iii) an Admin for the platform.

Functional Requirements

The use cases for the different user groups are listed below. Details of the use cases are given later. For each use case, its priority is also mentioned.

Name of the Use Case	Purpose	Precondition
For Students		
Explore upcoming Events (essential)	For users to explore upcoming events organized by various clubs	Student logged in
Explore clubs (essential)	For users to explore and obtain information about the various registered clubs.	Student logged in
Log in (essential)	To log in to the platform	None
Register for events (essential)	To register for events	Student logged in
View the profile (desirable)	Allow students to view and edit their profile	Student logged in
Propose a new club (essential)	Allow students to propose a new club	Student logged in

For Club Coordinator		
Propose an event (essential)	To propose a new event	Club coordinator logged in
Edit an event's information (essential)	Edit the event's information	Club coordinator logged in
For Admin		
Approve an event (essential)	Approve an event	Event has been created; admin logged in
Approve a new club (essential)	Events coordinator	Admin logged in
Change a club coordinator (optional)	Change the coordinator or their information	Admin logged in
Edit an event's information (optional)	Edit the event's information	Admin logged in

Performance Requirements

1. The system shall be designed to support up to 1000 concurrent users without degradation in performance.
2. The system shall have a response time of less than 300 milliseconds for 95% of the requests at maximum load.

Design Constraints

1. The institute's domain gmail address should be used for user authentication.
2. The tech stack for the platform should be free and open source.

External Interfaces

1. The platform should be compatible with all major browsers like Safari, Chrome, Firefox, Brave, Microsoft Edge, etc.
2. The platform should use users' institute email addresses for login and authentication.
3. The platform should be able to send email notifications to students and other users (Optional).

Security / Privacy Requirements

1. Only authorized users should be able to execute the different use cases; the landing page is visible to all.
2. The platform shall implement security measures including but not limited to input validation and output encoding (Optional).

Use Case Details

1. Explore Upcoming Events

1.1. **Actor:** Students

1.2. **Purpose:** For users to explore upcoming events organized by various clubs.

1.3. **Event Flow:**

1.3.1. The user lands on the homepage of the website.

1.3.2. The user views the list of upcoming events displayed on the main landing page.

1.3.3. If desired, the user can click on the "Home" button to refresh or return to the main landing page showcasing upcoming events.

1.4. **Special/Exceptional Requirements:**

None for this use case.

2. Explore Clubs

2.1. **Actor:** Students

2.2. **Purpose:** For users to explore and obtain information about the various registered clubs.

2.3. **Precondition:** Student is logged in

2.4. **Event Flow:**

2.4.1. The user clicks on the "Clubs" button on the navigation bar.

2.4.2. The user is redirected to the clubs' information page.

2.4.3. On this page, the user can view the club information for each club.

2.5. **Special/Exceptional Requirements:**

None for this use case.

3. User Log-In

3.1. **Actor:** Students, College Admin, Club Coordinators, Events Coordinator

3.2. **Purpose:** For students to log in to the platform.

3.3. **Event Flow:**

3.3.1. The user navigates to the main landing page of the portal.

3.3.2. The user clicks on the login button.

3.3.3. The user is redirected to log in through the Google sign-in API.

3.3.4. After a successful login, the user is redirected back to the main page; the login button is replaced with the logout button

3.4. **Special/Exceptional Requirements:**

None (google API will take care of incorrect logins)

4. Register for Events

4.1. **Actor:** Students

4.2. **Purpose:** To allow users to register for upcoming events.

4.3. **Precondition:** Student is logged in

4.4. **Event Flow:**

4.4.1. The user explores upcoming events from the main landing page.

4.4.2. The user clicks on an event to view its details.

- 4.4.3. The user clicks on the "Register" button.
 - 4.4.4. The user is registered and redirected to the homepage.
- 4.5. **Special/Exceptional Requirements:**
 - 4.5.1. If not logged in, the user is prompted to log in.
- 5. **View Profile**
 - 5.1. **Actor:** Students, Admin, Club Coordinators
 - 5.2. **Purpose:** To allow users to view their personal profile, which displays their details and the events they've registered for.
 - 5.3. **Event Flow:**
 - 5.3.1. From the main landing page, the user clicks on the "Profile" button in the navigation bar.
 - 5.3.2. Users, if logged in, are redirected to their respective profile pages.
 - 5.3.3. The user sees their name, roll number, email address, and profile type badge.
 - 5.3.4. The user can view the events they've previously registered for.
 - 5.4. **Special/Exceptional Requirements:**
 - 5.4.1. If not logged in, the user is prompted to log in.
 - 5.4.2. Profile-type badges are of the following kinds::
 - 5.4.2.1. Student: Regular students.
 - 5.4.2.2. Club Head: Club coordinators/heads for clubs.
 - 5.4.2.3. Admin: Administrative personnel.
- 6. **Propose an Event**
 - 6.1. **Actor:** Club Coordinators
 - 6.2. **Purpose:** To allow club coordinators to propose the organization of events.
 - 6.3. **Event Flow:**
 - 6.3.1. The user lands on the homepage and clicks on the "Clubs" option in the navigation bar, redirecting to the clubs' information page to locate and select their respective club from the list.
 - 6.3.2. On the club-specific page, the user clicks on the 'Propose Event' button, post which a form appears.
 - 6.3.3. After filling out this form, the user clicks the "Submit" button.
 - 6.3.4. The event proposal is sent for approval, and the user is redirected back to the club information page.
 - 6.4. **Special/Exceptional Requirements:**
 - None for this use case.
- 7. **Edit Clubs' Information**
 - 7.1. **Actor:** Club Coordinators, event coordinator, Admin
 - 7.2. **Purpose:** To allow specific roles to edit the information of the clubs.
 - 7.3. **Precondition:** Club coordinator/admin logged in
 - 7.4. **Event Flow:**
 - 7.4.1. **For a Club Coordinator:**

- 7.4.1.1. **The Club Coordinator/Head** selects their club from the "Clubs" page.
 - 7.4.1.2. The user clicks the 'Edit Club Information' button on the club-specific page.
 - 7.4.1.3. The user edits the club description and members list.
 - 7.4.1.4. The user clicks the "Save" button.
 - 7.4.2. **For the Admin:**
 - 7.4.2.1. The user chooses the 'Manage' option from the navigation bar.
 - 7.4.2.2. The user selects the 'Edit Club Information' from the management options.
 - 7.4.2.3. The user chooses the desired club and edits relevant fields.
 - 7.4.2.4. The user clicks the "Save" button.
 - 7.5. **Special/Exceptional Requirements:**
 - 7.5.1. Only the Club Coordinator/Head can see the club-specific 'Edit' button.
 - 7.5.2. The Events Coordinator and Admin access editing through the 'Manage' option.
- 8. **Approve Event**
 - 8.1. **Actor:** Admin
 - 8.2. **Purpose:** To evaluate and approve or reject event proposals from club coordinators/heads.
 - 8.3. **Event Flow:**
 - 8.3.1. The user selects the 'Manage' option after logging in.
 - 8.3.2. The admin chooses 'Pending Event Requests' from the available options.
 - 8.3.3. The user views a list of pending proposals and selects one to review its details.
 - 8.3.4. The user decides to 'Approve' or 'Reject' the event and clicks the appropriate button.
 - 8.4. **Special/Exceptional Requirements:**
None for this use case.
- 9. **Change a Club Coordinator**
 - 9.1. **Actor:** Admin
 - 9.2. **Purpose:** To change the details and assign the role of Club Coordinator to a new individual.
 - 9.3. **Event Flow:**
 - 9.3.1. The user logs in and selects the 'Manage' option from the navigation bar and chooses the 'Update club Coordinator' option.
 - 9.3.2. The user views the current Coordinator's details.
 - 9.3.3. The user can click on "edit" and update the required details.
 - 9.3.4. The user can save the changes, and update access and profile badges for the current and new coordinators.
- 10. **Request the Formation of a New Club**

- 10.1. **Actor:** Students
- 10.2. **Purpose:** To propose the formation of a new club.
- 10.3. **Event Flow:**
 - 10.3.1. The user selects the 'Clubs' option on the homepage and lands on the clubs' information page.
 - 10.3.2. The user clicks on 'Propose a New Club' and fills out the form.
 - 10.3.3. The user submits the form, sending the request forward for approval.
- 10.4. **Special/Exceptional Requirements:**
 - None for this use case.

11. Approve New Club Requests

- 11.1. **Actor:** Admin
- 11.2. **Purpose:** To evaluate and approve/reject new club formation proposals.
- 11.3. **Event Flow:**
 - 11.3.1. The user logs in, selects 'Manage', and navigates to 'Pending New Club Requests'.
 - 11.3.2. The user reviews and decides to 'Approve' or 'Reject' a specific club proposal. Approved proposals are made available for Admin review.
 - 11.3.3. The user reviews and finalizes the decision by either approving it, thereby adding the new club to the portal, or rejecting it.
- 11.4. **Special/Exceptional Requirements:**
 - 11.4.1. Only proposals approved by the Events Coordinator are visible to the Admin.

Functional Requirements in terms of Features/Functionality.

Functional requirements can also be stated directly as functionality or features the application should support. This section specifies the functionality in terms of features.

1. The system shall allow users with an active domain email address of the institute for various functionalities. The authentication process shall verify the domain of the email address to ensure it belongs to the institute. Users without a valid institute domain email address shall receive a specific error message indicating the reason for authentication failure. (all)
2. The platform should enable the club coordinators to schedule events for any given date and time. While applying for the same, they should be permitted to edit the event details and registration requirements. (Club Coordinators)
3. The platform would also have dedicated pages for the clubs, which the respective club's coordinator would manage. A club coordinator should be allowed to edit this page by editing club information, images from past events, and point of contact information. (Club coordinators)
4. The admin should have access to a page that lists all pending requests for scheduling the events while marking potential conflicts and should be able to approve or decline each one of them. (admin)

5. After the event's approval, it should be added to the platform for registration by students of the institute, and an email notification should be sent to all the students. (Optional) (all)
6. Students should be allowed to search for upcoming events based on their interests and view them. (all)
7. The platform's admin access should be given to an admin who can edit event details, Club Coordinator and respective clubs' and students' accounts information. (Admin)
8. There should be a report feature in the platform to enable everyone to report incidents and issues with the platform. (all)
9. Students should be able to request for the formation of a new club. (all).
10. The approval for new clubs will be given by the admin. (Admin)

Glossary of Terms:

1. Students: Users who study at the Institute
 2. Club coordinators: Students who are responsible for managing the club activities and hosting events.
 3. Events coordinator: They are student users responsible for managing and approving the events added by club coordinators/heads.
 4. Admin: Appointed by the institute and is responsible for approving events, ensuring their management, etc. can change logins and credentials, and manage the application.
 5. Credentials: information required for logging in; in this case, it is the Google account associated with the institute.
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